AI ACADEMY · LEVEL 3 · GRADE 8 · WEEK 3 · STUDENT LESSON

Tokens and Context

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

Reserve the space. Keep the qualifier. Check the claim.

Budget context while preserving current complete evidence and explicit token conventions.

Keep one habit: Toy accounting → source selection → exception case → tokenizer/card audit → independent budget.

MAIN LESSON · SECTION 1

Which evidence must fit for this answer to be supported?

> Budget supplied token units without dropping decisive qualifiers.

Your learning goals

I can account for fixed text, **evidence** and **reserved output** under a toy budget.

I can select complete **evidence** for a stated question.

I can handle insufficient or conflicting **evidence** explicitly.

Recall probability versus factual support

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Does lower next-**token** loss guarantee supported factual answers?

Your end goal

By the end, I can budget context and preserve **evidence** needed to interpret a requested fact.

▼ Attention extension · Equal sums can hide order

Why can equal row sums conceal different sequences?

> **My readiness explanation**

▼ Earlier wording, alternatives and picture transcripts

Your learning target

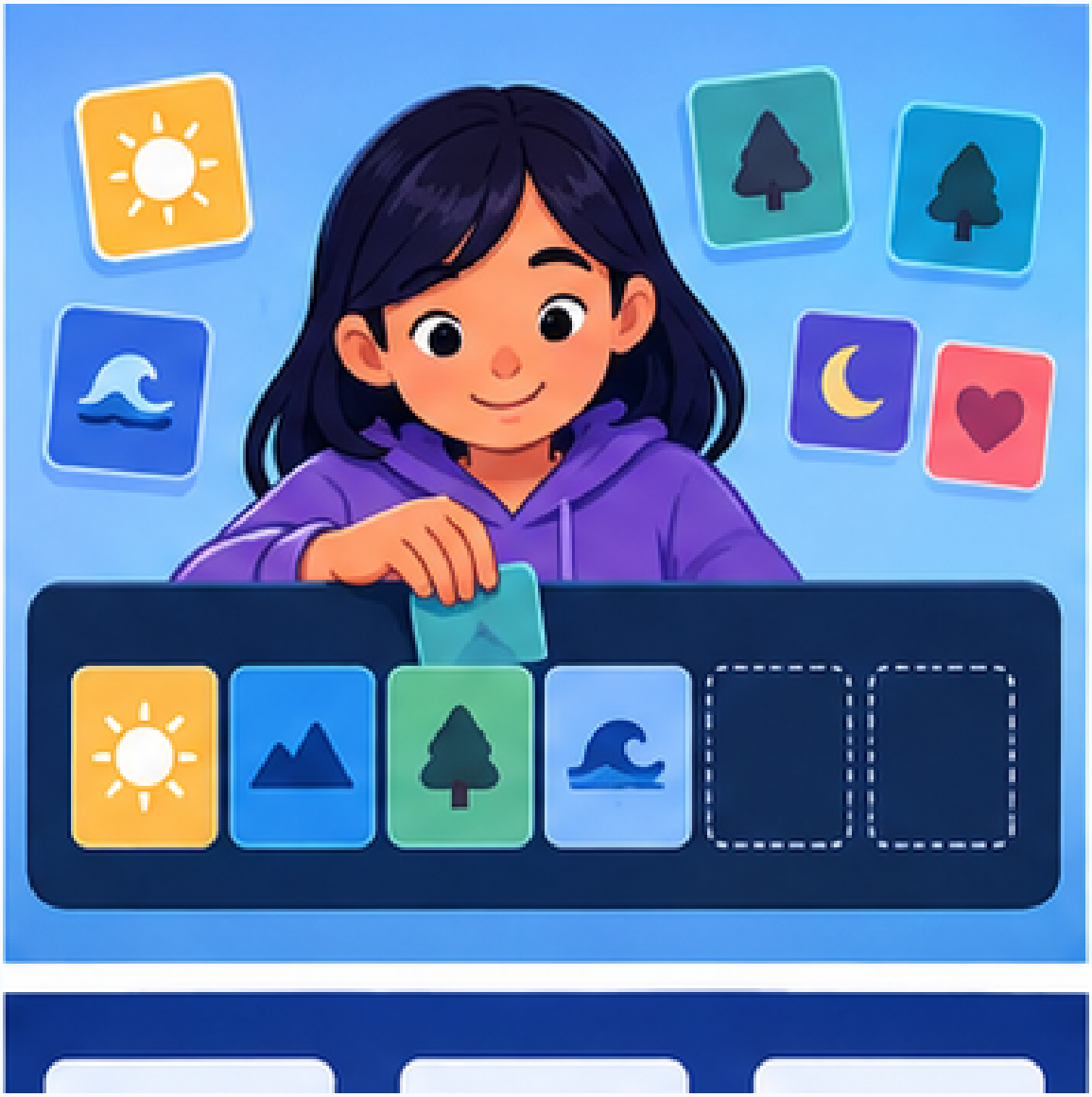
Remember before reading

Without opening the old lesson, explain one key idea from A Distribution of Possibilities. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 3: **Token**s and Context

Models operate on **token**s within a limited **context window**.



CREATOR MISSION

Tokenize text, remove context, and predict failure changes.

Creator name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for Attention as weighted information selection

This week’s end goal

MAIN LESSON · SECTION 2

Reserve space before selecting evidence

> The toy accounting rule is supplied; it is not a deployed model limit.

The main100-unit budget

For this lesson only, a fictional system allows 100 total **token** units for **input** plus **reserved output**. Its fixed instructions use 20 units, question 10 and **output** reserve 20. That leaves 50 units for **evidence**.

These are supplied counts, not word counts or a claim about any deployed model. Other systems can use different **input**/**output** accounting, so read their exact contract.

The room-approval question and complete evidence facts

The question is: “Which room is approved for the robotics meeting, and what is its capacity?” **Evidence** A, a current approval notice, costs 30 units and names room B.

Evidence B, the current room register, costs 20 and gives room B capacity 12. **Evidence** C, an old approval notice, costs 25 and names room A.

▼ **Attention extension · Supplied weights and values**

Attention computes a weighted combination of **value** vectors, with weights derived from **query**-key compatibility. Queries, keys and **value**s are distinct roles even when they come from the same sequence. A weight is a numerical contribution in this computation, not proof of the source’s truth or permission. This week supplies weights directly; Week4’s retained attention extension calculates them from scores.

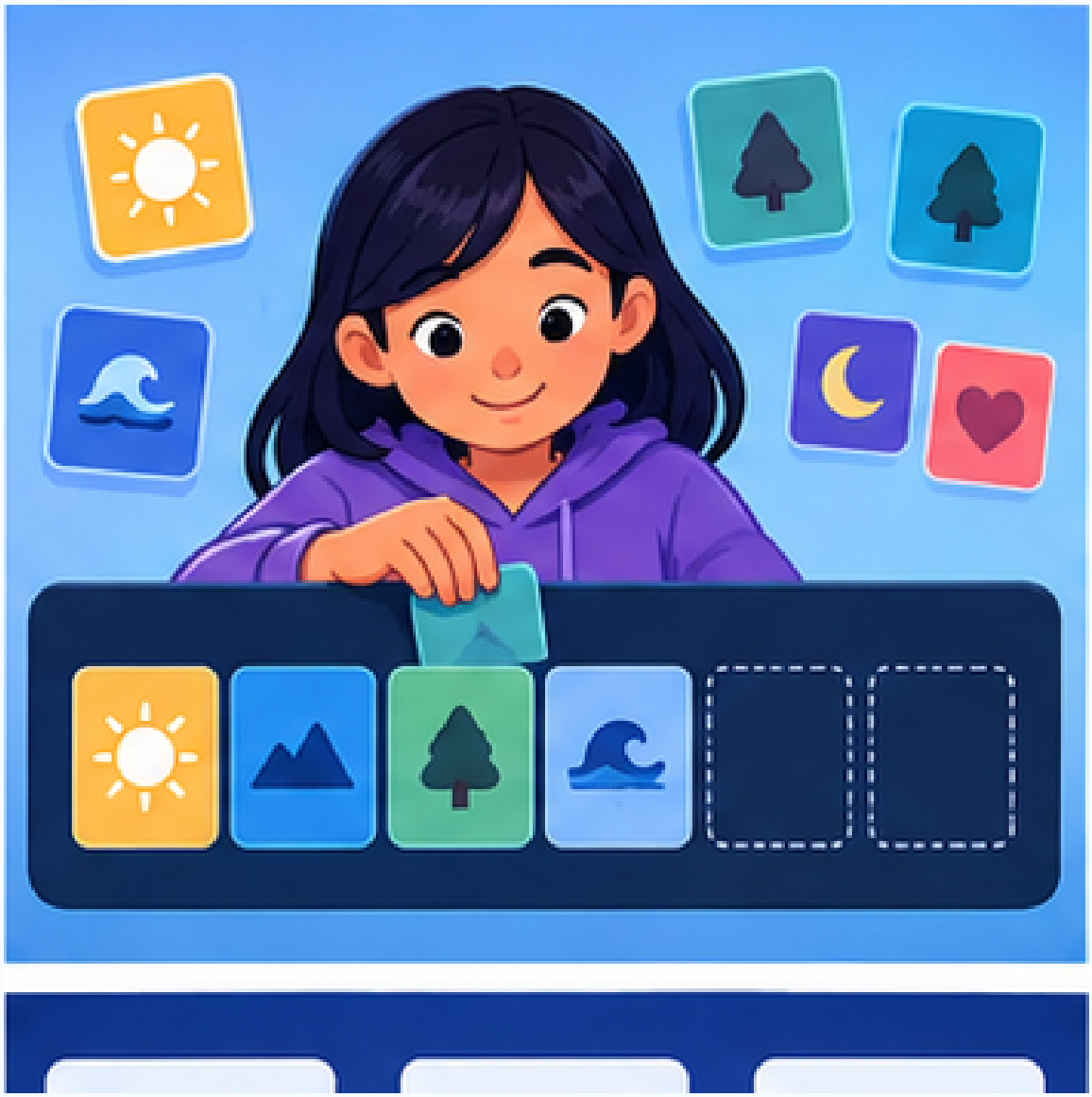
Use three **value**s V1=[2,0], V2=[0,4], V3=[4,2] with weights 0.5,0.25,0.25. **Value**s have equal width and weights are nonnegative with sum 1. Compute each coordinate separately. No **token** is literally copied by this combination.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **token**s and context to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: Attention as weighted information selection

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MAIN LESSON · SECTION 3

Check both size and support

> An old approval can fit without establishing current approval.

Read the complete room evidence table

Evidence	Cost	What it supports
A: current approval	30	Approved room B
B: current register	20	Room B capacity 12
C: superseded approval	25	Old room A approval

Compare all three evidence pairs

A+B costs 50 and fits exactly. Together they support both requested claims. A+C costs 55 and exceeds the **evidence** budget, while C+B fits at 45 but supplies an outdated room decision.

A cheap set is not necessarily an adequate set. Preserve document IDs, versions and dates alongside content.

Trace the50-unit evidence allocation

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
100 total units	Subtract 20+10+20	50 evidence units
A 30 plus B 20	Exactly 50 evidence units	Supports both requested claims
Insufficient evidence	Ask or abstain	Do not invent the missing fact

Arithmetic fit and evidential adequacy are separate checks.

Explain the deciding evidence

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Attention extension · Coordinate-wise weighted combination

Value	Weight	Weighted vector
[2,0]	0.5	[1,0]
[0,4]	0.25	[0,1]
[4,2]	0.25	[1,0.5]

Adding the weighted vectors gives [2,1.5]. The same weights applied to different **value**s can give a different result; the same **value**s with different weights can also change it. For example weights 0,1,0 yield V2=[0,4].

The result is a representation to be used by later operations, not automatically a generated word.

Keep the evidence roles separate

PLAN OR INPUT	PROCEDURE	PERMITTED CONCLUSION
Values with equal width	Multiply by .5, .25, .25	Contributions [1,0], [0,1], [1,.5]
Weighted contributions	Add corresponding coordinates	Output [2,1.5]
Mask excludes V3	Renormalise V1 and V2	Output [4/3,4/3]

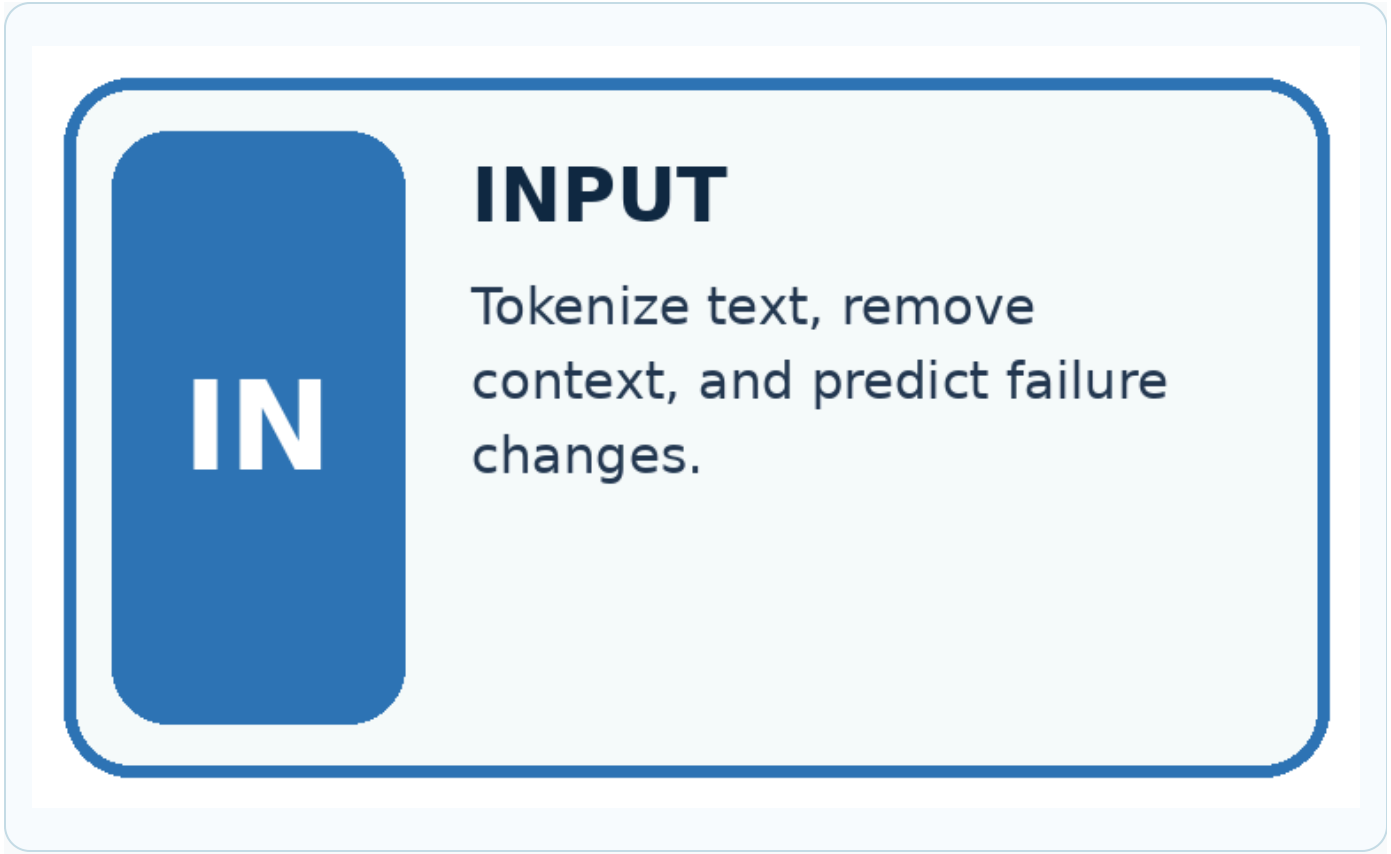
The mask changes the eligible positions and therefore the probability mass assigned to the remaining values.

▼ Earlier wording, alternatives and picture transcripts

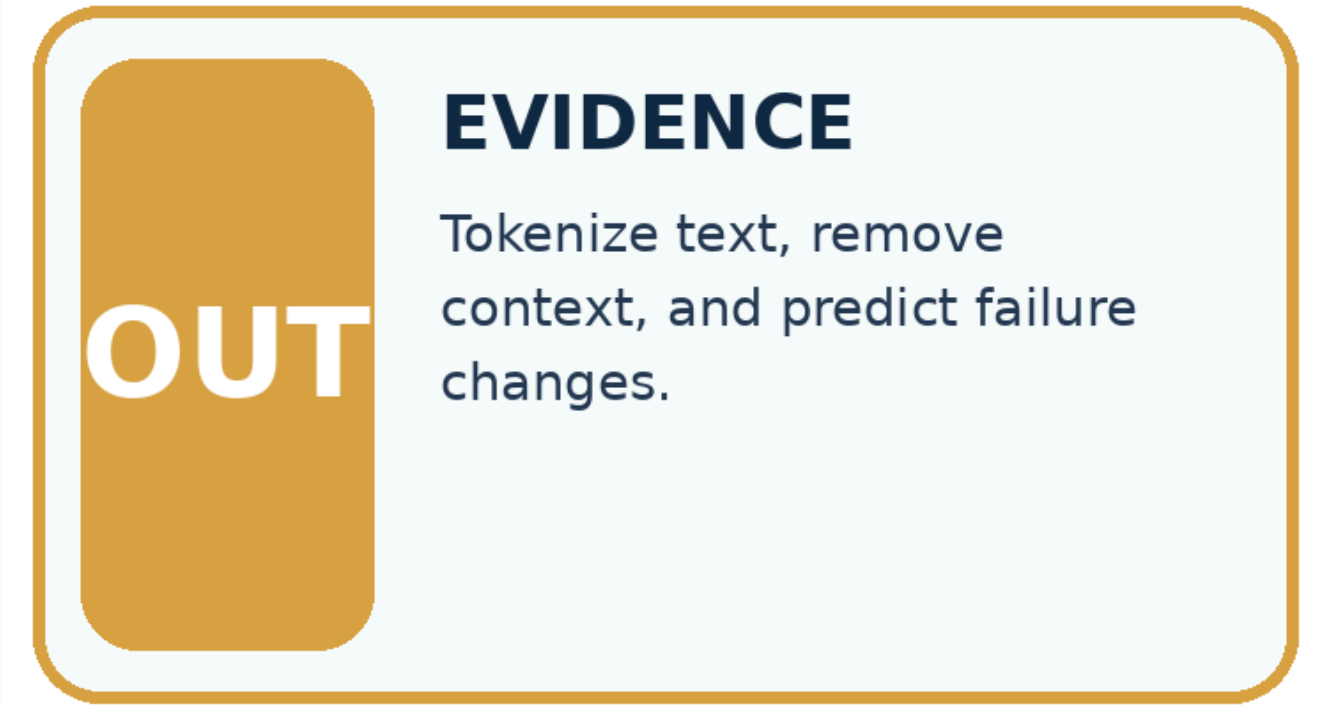
VISUAL **MECHANISM**

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Tokenize text, remove context, and predict failure changes.
↓
SYSTEM
Models operate on tokens within a limited context window.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

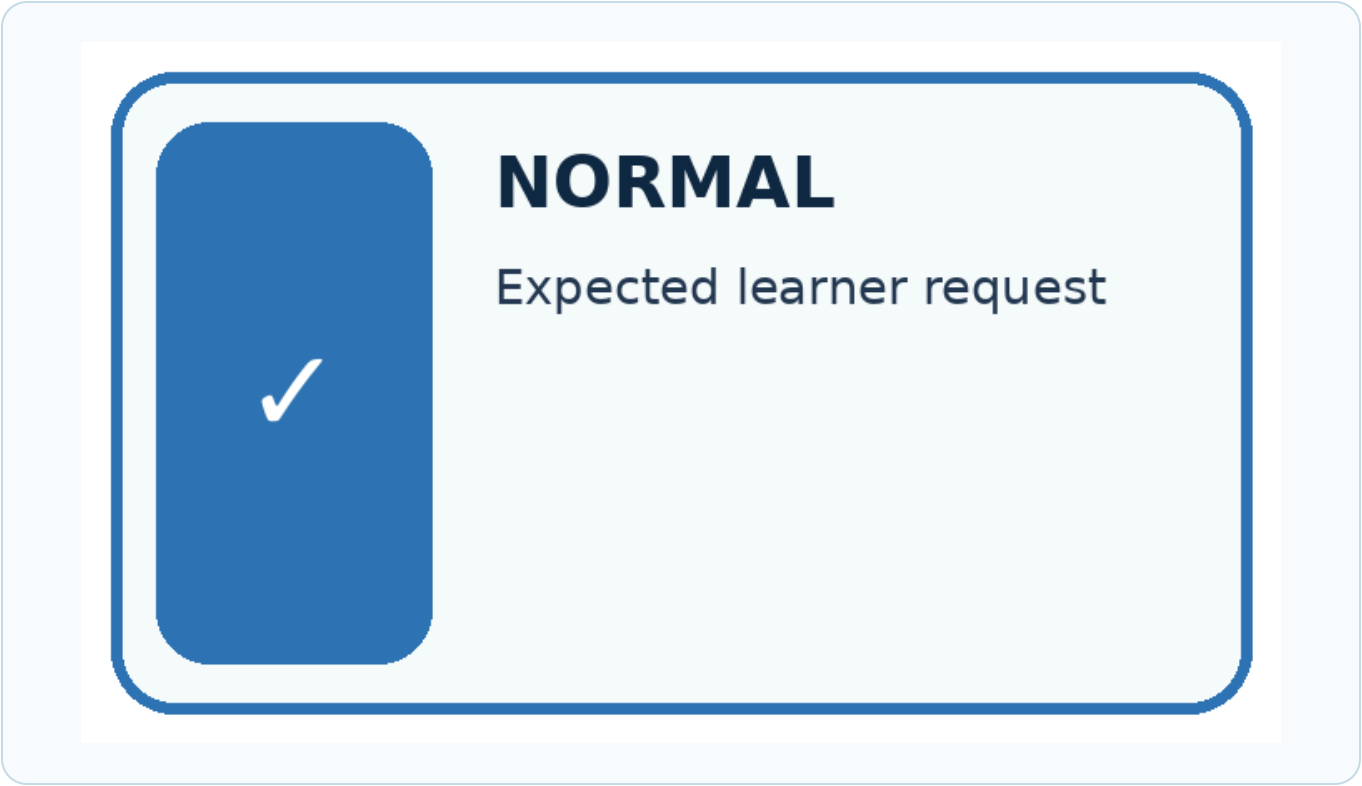
Apply the connected idea: Attention as weighted information selection

Connect the picture and the explanation

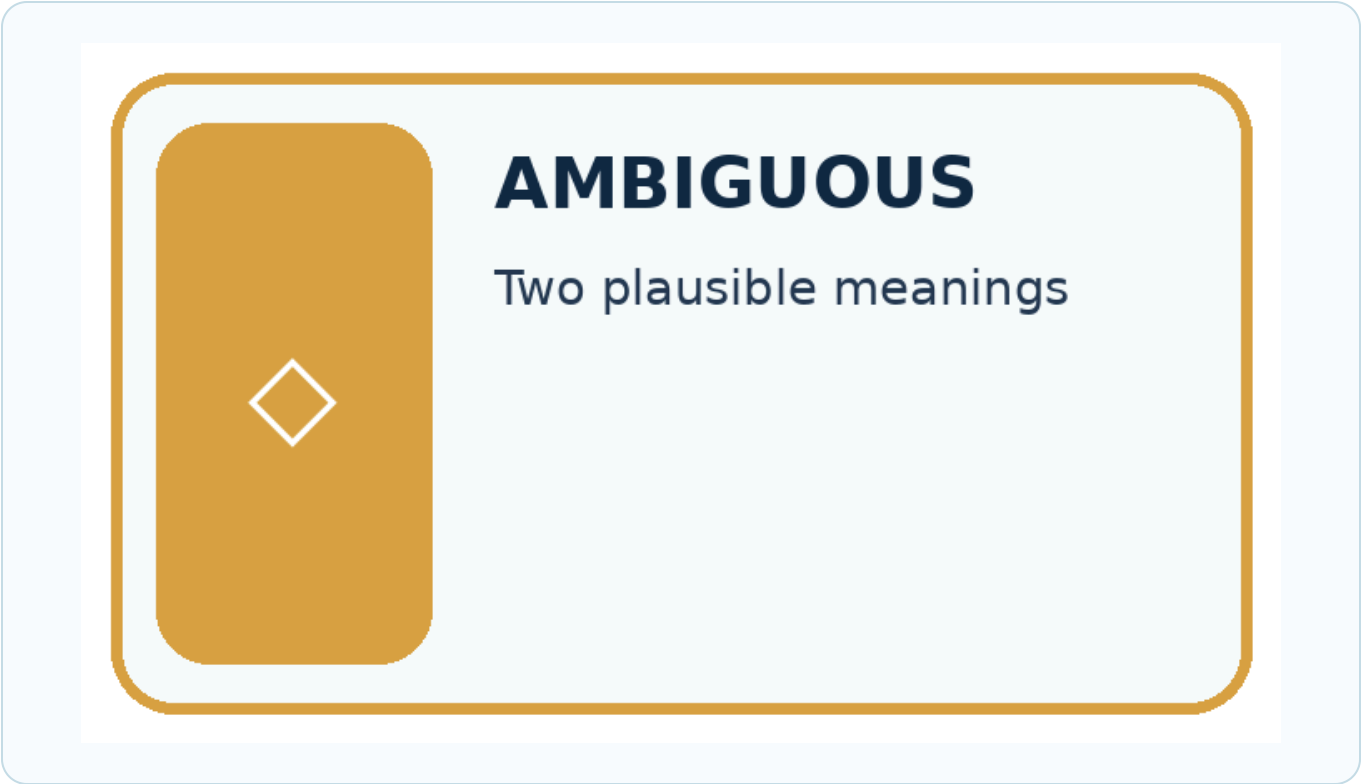
Compare complete evidence with a cheap incomplete shortcut

> A smaller context is useful only if it preserves the needed meaning.

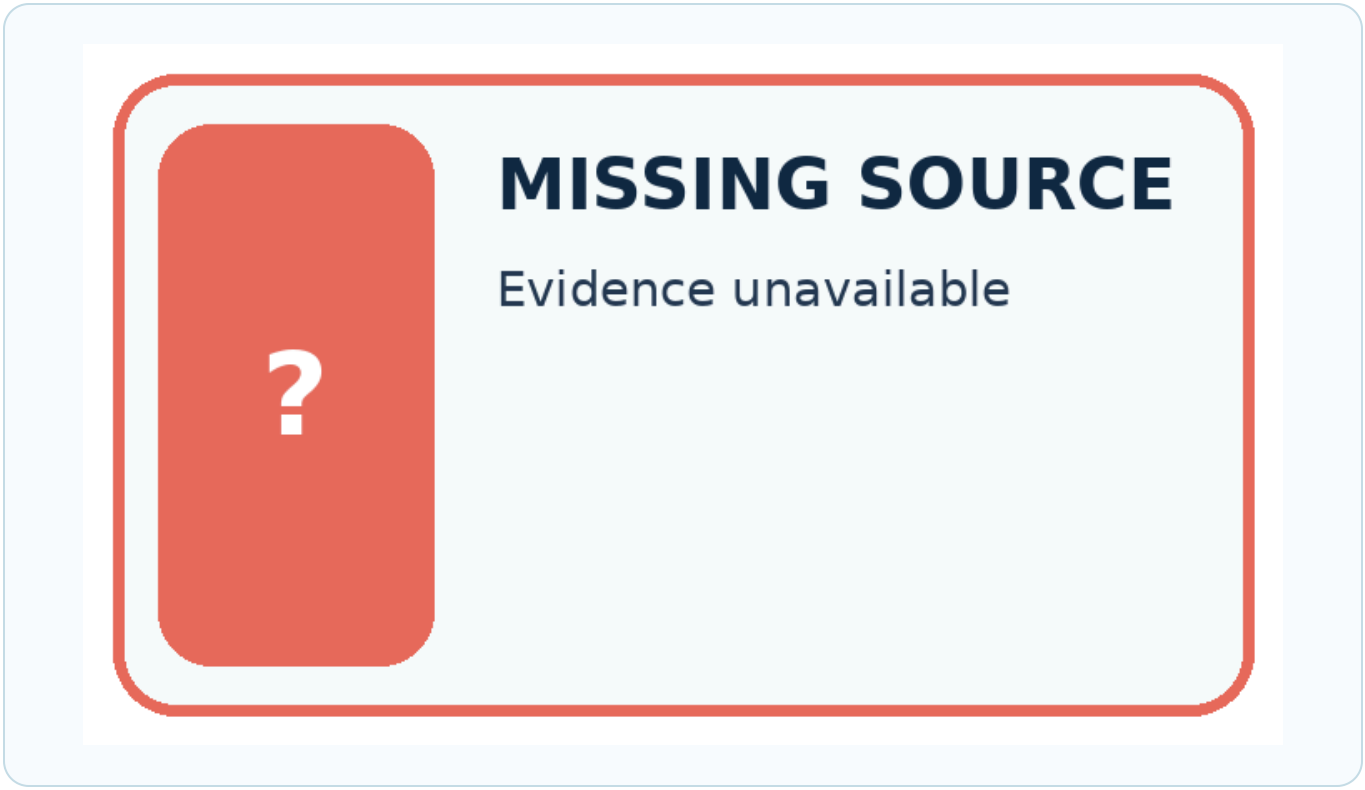
Four pictures with a purpose



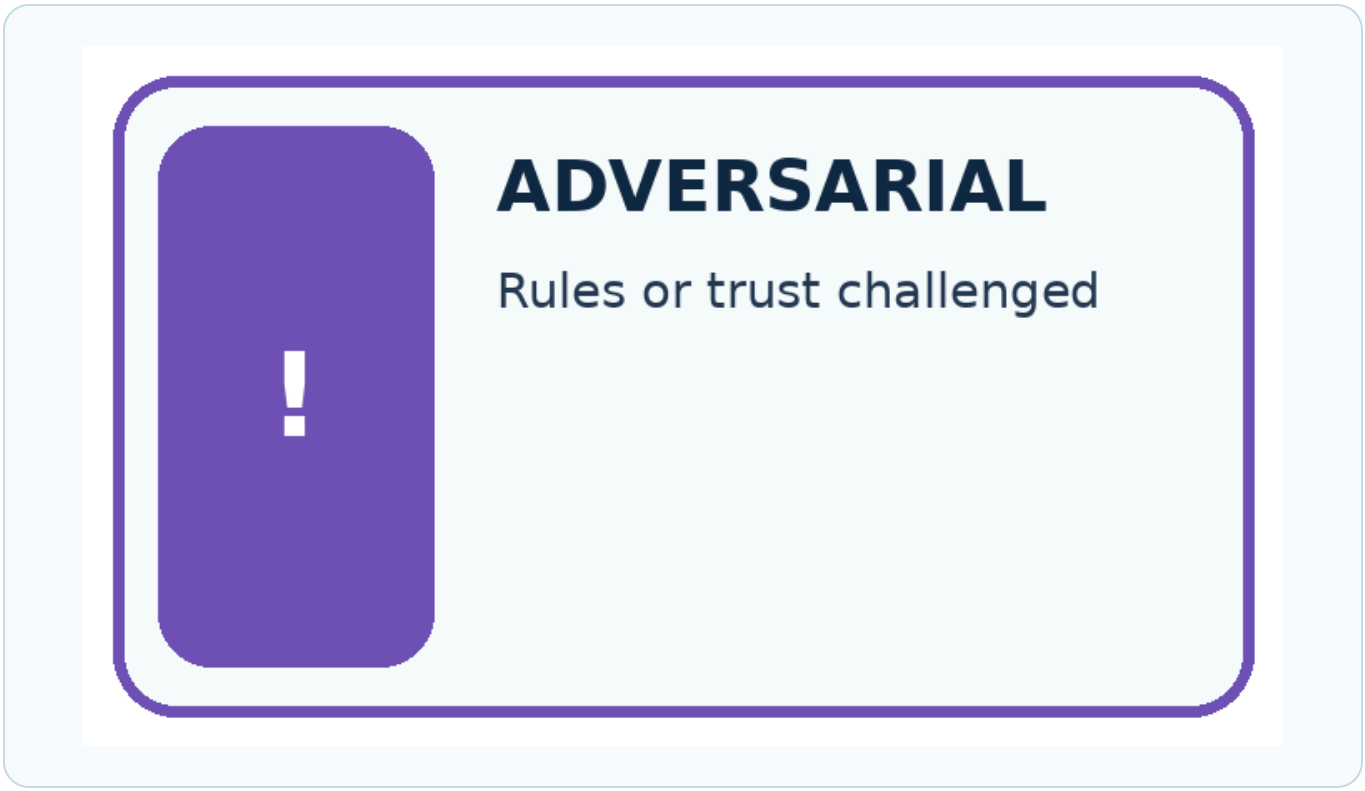
Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Increase B's cost to 30 with all other costs fixed.

▼ Check after predicting

Prediction: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

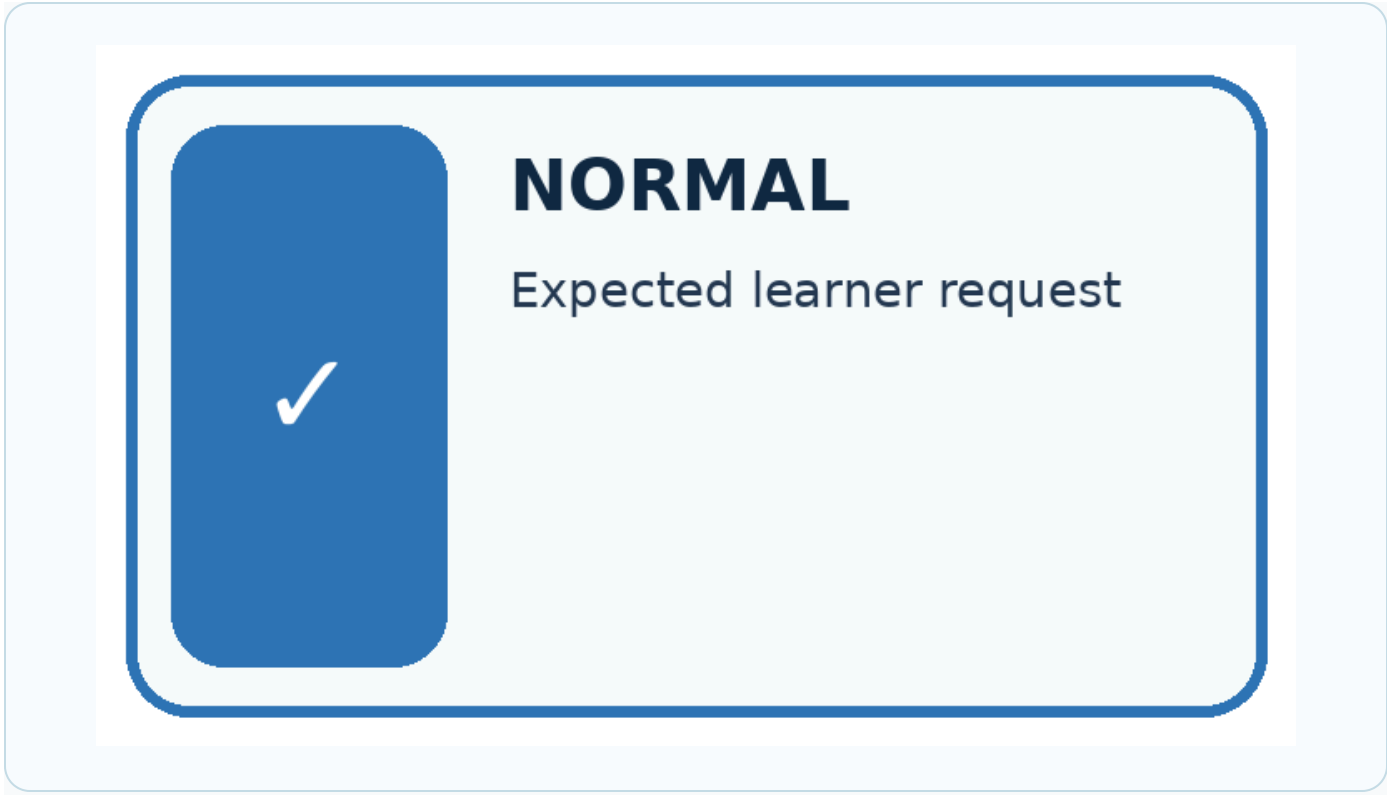
Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

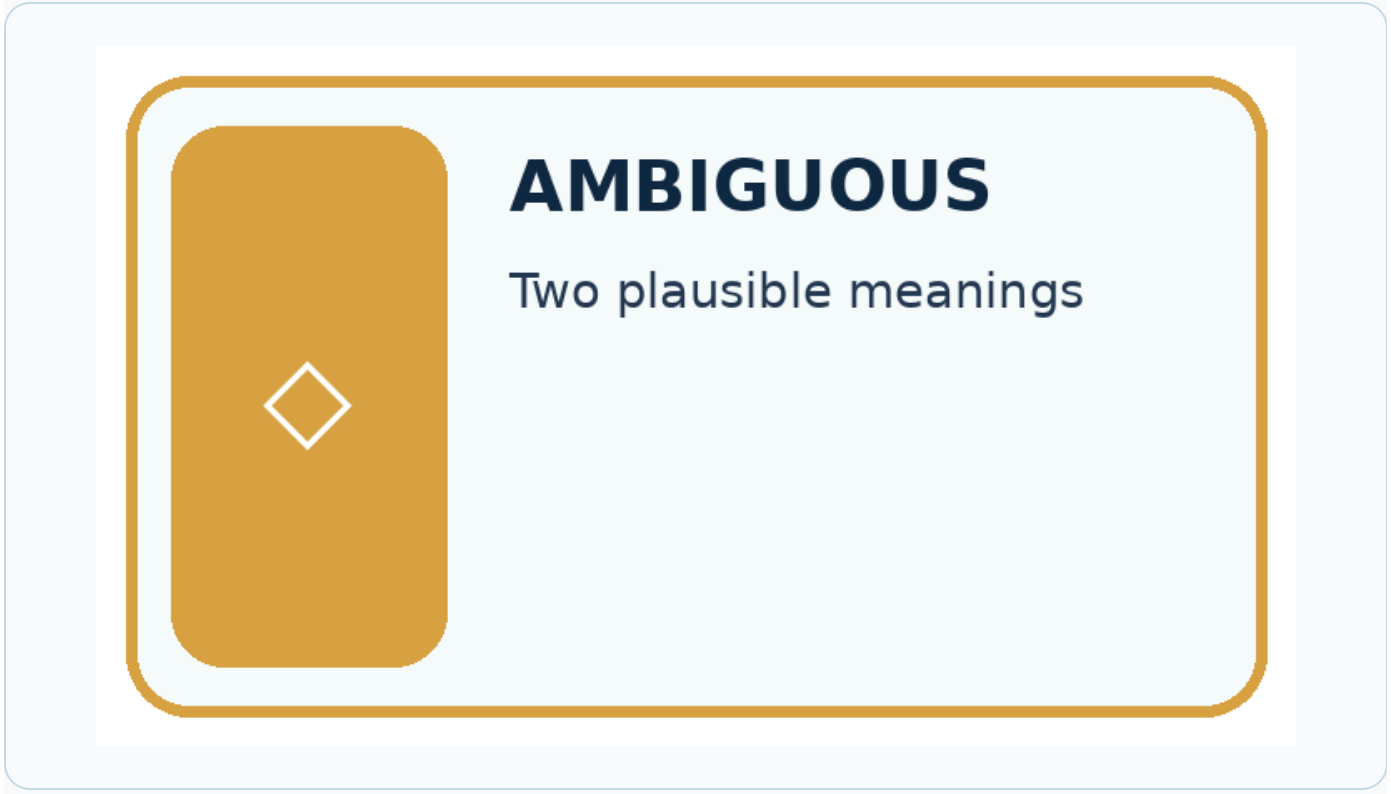
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Preserve an exception in a separate event-capacity case

> The main approval example reserves20output units; this picture reserves25.

Keep qualifiers, versions and source authority

A source may contain an exception after a general statement. Cutting off the last paragraph can reverse its practical meaning.

Selection should preserve the passages needed to interpret the requested fact, including scope, version and exceptions.

If complete relevant **evidence** cannot fit, retrieve a smaller authoritative passage or request clarification rather than silently remove a qualifying condition.

Including a document in context does not prove the generated answer uses it correctly. After generation, check each claim against its source.

A source’s instruction-like content also does not acquire authority just because it fits in the **evidence** budget.

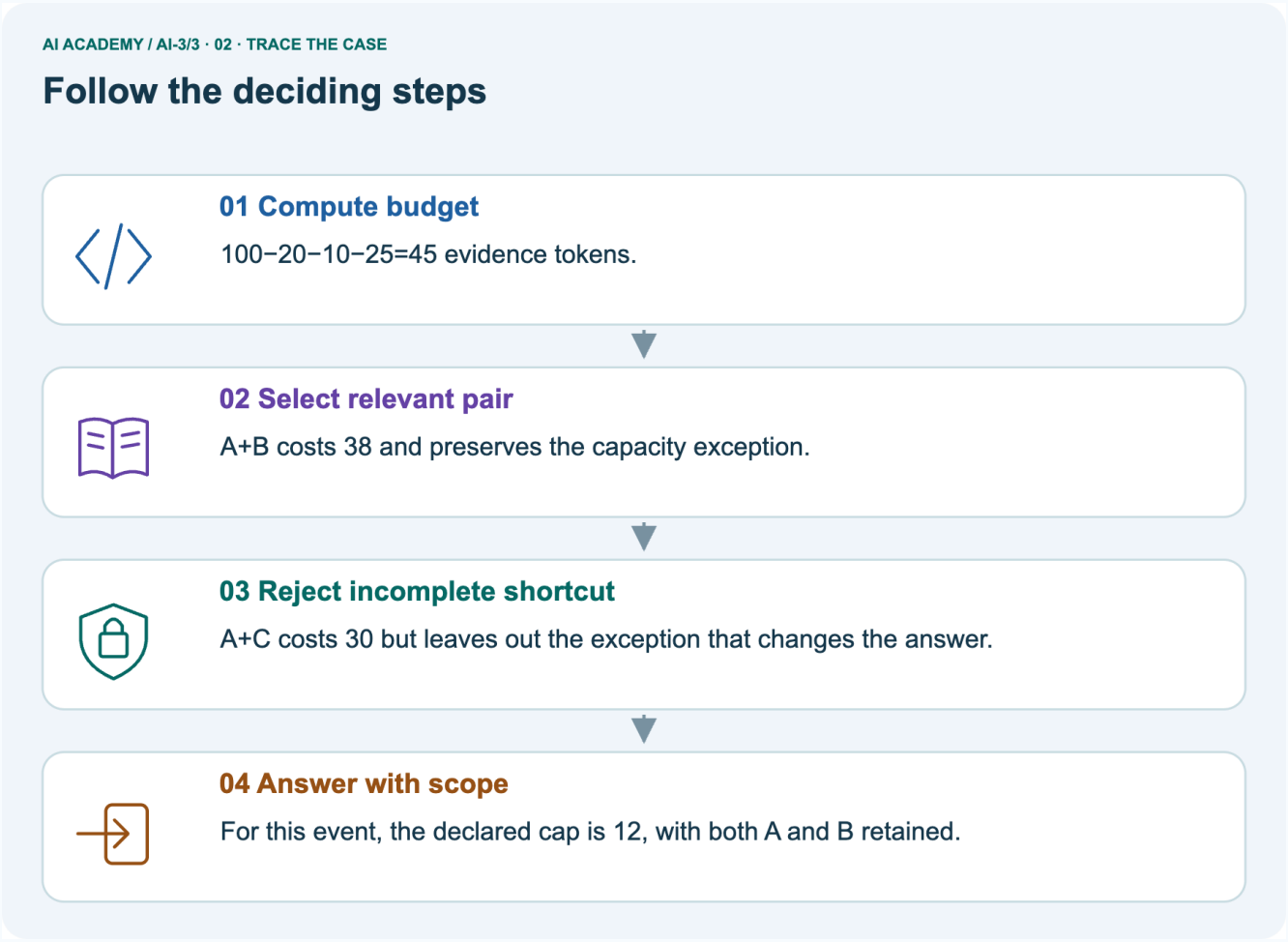
Read the separate event budget and records

A fictional assistant has a small **context budget** for answering an event-capacity question.

Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Trace the relevant pair and scoped answer



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute budget	100−20−10−25=45 evidence tokens.
2. Select relevant pair	A+B costs 38 and preserves the capacity exception.
3. Reject incomplete shortcut	A+C costs 30 but leaves out the exception that changes the answer.
4. Answer with scope	For this event, the declared cap is 12, with both A and B retained.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Predict the larger exception passage

- Change one thing: Increase B's cost to 30 with all other costs fixed. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting

AI ACADEMY / AI-3/3 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

For this event, the declared cap is 12, with both A and B retained.

Changed condition

Increase B's cost to 30 with all other costs fixed.

Worked changed-case result

A+B costs 48 and no longer fits the 45-token evidence capacity.
Why: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-evidence response.

Follow the labelled picture with the matching explanation. Open the vector diagram

Explain fit versus faithful use

- **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.
- Read the labelled picture: Can A alone justify answering 20 seats after the exception is known?

▼ Attention extension · Mask, renormalise and handle no available value

Suppose V3 is a future position unavailable to the current autoregressive step. The **mask** excludes it. For this supplied-weight example, renormalise the permitted weights 0.5 and 0.25 by their total 0.75, giving 2/3 and 1/3.

The result becomes [4/3,4/3]. Merely setting V3′s contribution to zero while keeping the old weights produces [1,1], whose retained weights sum to 0.75; that is not the specified normalised combination.

If every position is excluded, the permitted total is zero. This contract reports no available **value** rather than dividing by zero or silently using a future position.

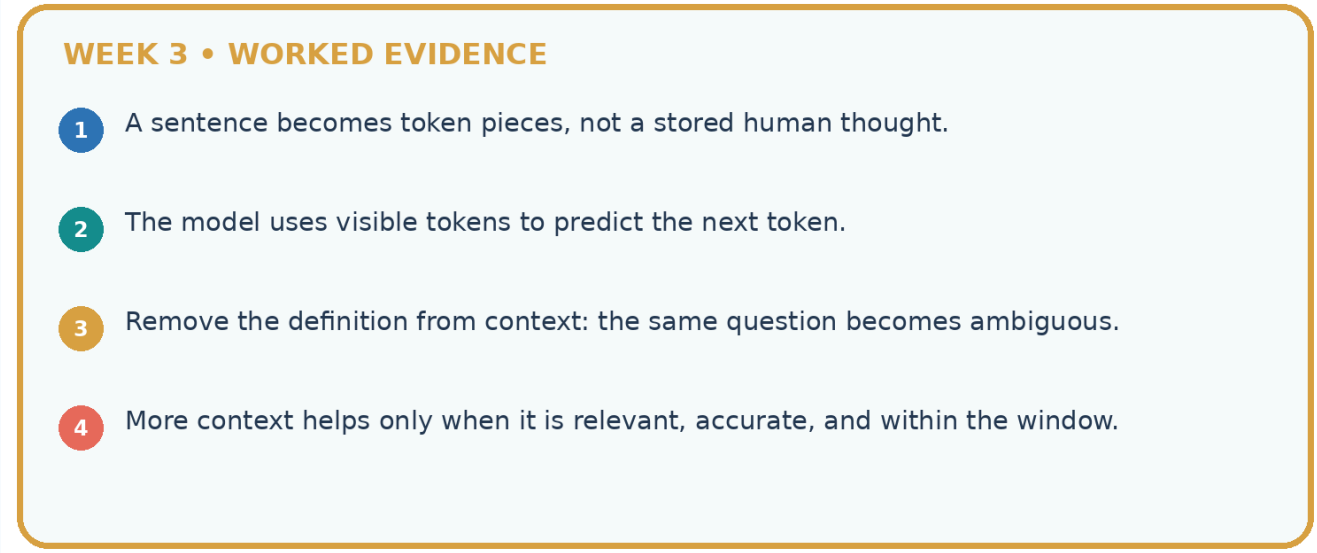
A real implementation needs its own documented handling for empty **mask**s. A causal **mask** restricts information paths; it does not make retrieved content factually trustworthy.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. A sentence becomes **token** pieces, not a stored human thought.

2. The model uses visible **token**s to predict the next **token**.

3. Remove the definition from context: the same question becomes ambiguous.

4. More context helps only when it is relevant, accurate, and within the window.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Apply the connected idea: Attention as weighted information selection

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Tokens and Context

The facts we will use

AI ACADEMY / AI-3/3

Context budget allocations

	Toy tokens
Instructions	20
Question	10
Output reserve	25
Evidence available	45

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Find the meaning lost by a shorter selection

> Budget compliance and answer support are separate checks.

Repair the shortest-pair argument

Claim to repair: A+C is shorter than A+B, so it is better evidence for the event. In the pictured event case A says20seats, B states this event's cap12 and C is unrelated. A+C fits at30units but omits the decisive exception. Preserve A+B at38 within45. If B grows to30, their48-unit pair no longer fits; request an approved source-preserving smaller passage or report incomplete evidence.

Do not knowingly answer20 for the event after seeing its exception.

Design a test that changes the evidence need

> **Exit: A is unavailable and B gives only room capacity. Write a response that reports the supported fact without claiming the meeting is approved there.**

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A token is always exactly one English word. Strong move: inspect the claim, source, stage, schema, or test evidence; repair one cause; rerun.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

If room approval is missing, do not infer it from capacity. If sources conflict without a documented current version, report the conflict or ask the responsible person.

Keep the requested claim, budget calculation, selected source IDs and unresolved facts in the record. This classroom accounting develops the reasoning without needing a live model.

Apply the connected idea: Attention as weighted information selection

A large weight means a large coefficient in this particular weighted sum. It is not a complete explanation of the whole network, which may contain other heads, transformations and residual paths.

It also does not establish that the high-weight source is true. Trace the actual **value**s and downstream use before making a broader claim.

> **Exit: a team says a 0.9 attention weight proves a document is correct and allows following its instructions. Explain the two unsupported inferences.**

MAIN LESSON · SECTION 8

Practise a separate80-unit budget with less help

> Use the accounting rule for the current case.

Read the complete80-unit record

Supported total 80, fixed instructions 15, question 10 and **reserved output** 15 leave 40 units. Two complete relevant passages cost 18 and 22, **fitting** exactly.

A third irrelevant passage costs 8; adding it would exceed the stated total. A short summary can reduce size but must retain claim-supporting qualifiers and be checked against the original.

Compute fit and explain its limit

> **Guided: calculate the available evidence units and explain why exact fit does not prove answer correctness.**
>
> _____
>
> _____
>
> _____
>
> _____

Complete main room-approval W2

Practise with less help: Choose complete support

Use workbook W2 for the connected example “Context limits and **evidence selection**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Repair token-count equals word-count

- Use the worked case for Context limits and **evidence selection**. Proposed explanation: “**Token** count equals word count”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Check your reasoning

Explain the separate event exception

- Can A alone justify answering 20 seats after the exception is known?

Explain the picture

> **My explanation using a pictured step and the given facts**

Keep independent work separate

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Attention extension · Compare a masked information set

Supported **value**s A=[6,0], B=[0,3], weights 1/3 and 2/3. Their weighted combination is [2,2]. **Mask**ing B leaves A with normalised weight 1 and result [6,0].

This is a different information set, not a numerical repair to the original result.

> **Guided: compute both results and explain why a masked-out source must not influence the combination.**

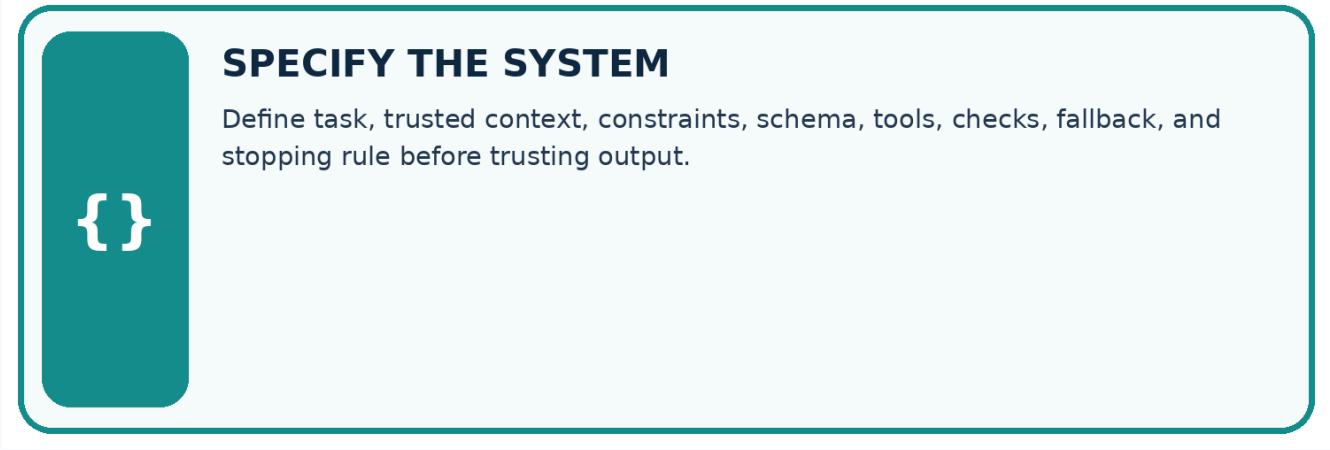
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: Attention copies the highest-weight **token**

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Specify Before You Run

Prompts, schemas, retrieval, tool calls, and code are inspectable parts of one AI product.



SET task, audience, constraints, **output**_schema LOAD approved_context or retrieval_result CALL model or allowed_tool VALIDATE fields, claims, citations, and safety IF **evidence** missing: ask, retrieve, or abstain RETURN useful **output** + source trail + limitations

Trace one **input**. Which line changes the state or chooses a path?

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: Attention as weighted information selection

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Practise the connected topic: Attention as weighted information selection

Tokenise with an explicit rule, then audit complete context cards

> Keep vocabulary results, evidence selection and actual model behavior distinct.

Predict and run the complete toy encoder

```
vocabulary = {1:'Ada',2:'AI',3:' ',4:'!',5:'🙂',6:'school',7:'book',8:'schoolbook'}

def encode(text):
    ids, position = [], 0
    while position < len(text):
        matches = [(i,piece) for i,piece in vocabulary.items() if
text.startswith(piece,position)]
        if not matches:
            return ids, (position,text[position])
        token,piece = max(matches,key=lambda pair:len(pair[1]))
        ids.append(token)
        position += len(piece)
    return ids,None

for text in ['Ada!','AI🙂','schoolbook','school book','ada']:
    ids,error = encode(text)
    print(repr(text),ids,error)
    if error is None:
        assert ''.join(vocabulary[i] for i in ids)==text
```

State its vocabulary and failure contract

1. Toy tokenisation investigation: the complete local program supplies its case-sensitive vocabulary and longest-match rule. Predict the token IDs for each of the five strings, then run or make a labelled paper trace. Spaces, punctuation and the emoji are explicit entries; schoolbook has one longest matching token here. The lowercase name fails at position0 rather than being silently changed.

   Error positions are Python string positions, not byte offsets. This toy vocabulary does not describe any commercial tokenizer, and no special end token is appended.

Trace evidence selection with full supplied cards

1. Separate evidence-card investigation: use the complete main room records A(current approval roomB,cost30), B(current register roomBcapacity12,cost20), C(superseded approval roomA,cost25). Available evidence capacity is50. Compare A+B, A+C and B+C for size and support of the two requested facts. Hide A, then hide B: record which fact cannot now be supported.

   Do not invent an actual model continuation or assume unseen evidence was used. Next trace the separate event picture: its45-unit budget and12-seat exception use different A/B/C labels. Keep the independent D/E/F evidence-budget task withheld.

Keep the evidence record

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → RUN → VERIFY → EXPLAIN

Tokenize names, punctuation, emojis, and compound words; then hide different context cards and compare predicted continuations.

Use the supplied toy vocabulary/encoder and complete room-evidence cards. Predict exact **token** IDs, preserve unsupported-character failures, then compare size and claim support when a card is hidden. These are local calculations and paper traces, not measured model continuations.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: budget the fresh D/E/F evidence

> Keep source currency, completeness and token cost separate.

Independent record and scope

Total 120, instructions 25, question 15, **output** reserve 30. Source D is current approval, cost 28; E is current capacity, cost 22; F is old approval, cost 18.

Compute capacity, compare D+E and E+F, and state an honest fallback if D is missing.

The independent task supplies source roles and costs, not actual room values. Compute fit and explain which evidence role is missing; do not invent an approved room or capacity. If D is missing, distinguish a capacity fact supported by E from the unresolved approval needed to link the response to the current event.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Budget new evidence independently

Week 3 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

A system can use only the last two messages. Message 1 says use centimetres; messages 2 and 3 discuss plants. What instruction may be lost, and how can the next request repair this?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Keep the learner's question, approved source excerpt, and **output** rules inside the active context.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

A toy system allows 12 **token**s total for **input** and **reserved output**. Instructions use 3, a source uses 7 and the planned answer needs 4. Compute the excess and propose a revision that preserves **evidence**.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Preserve the source relationship inside the budget

> A context plan must retain the evidence needed for each requested claim.

This week’s concrete contribution

For the Verified Study Buddy, keep instructions, learner question, approved evidence, source IDs/versions, relevant exceptions and reserved response space within a stated accounting contract. Use actual tokenizer counts only when a specific tokenizer and convention are supplied; otherwise label counts as toy units. If complete support cannot fit, seek a checked smaller passage or state the gap.

After generation, check each claim against the retained source; fit alone is not verification.

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**.

Keep the checkpoint and human choice

THIS WEEK’S CHECKPOINT Keep the learner’s question, approved source excerpt, and **output** rules inside the active context.

SYSTEM MAP Learner need → prompt/schema → approved sources/retrieval → model/tool → structured answer → verification → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

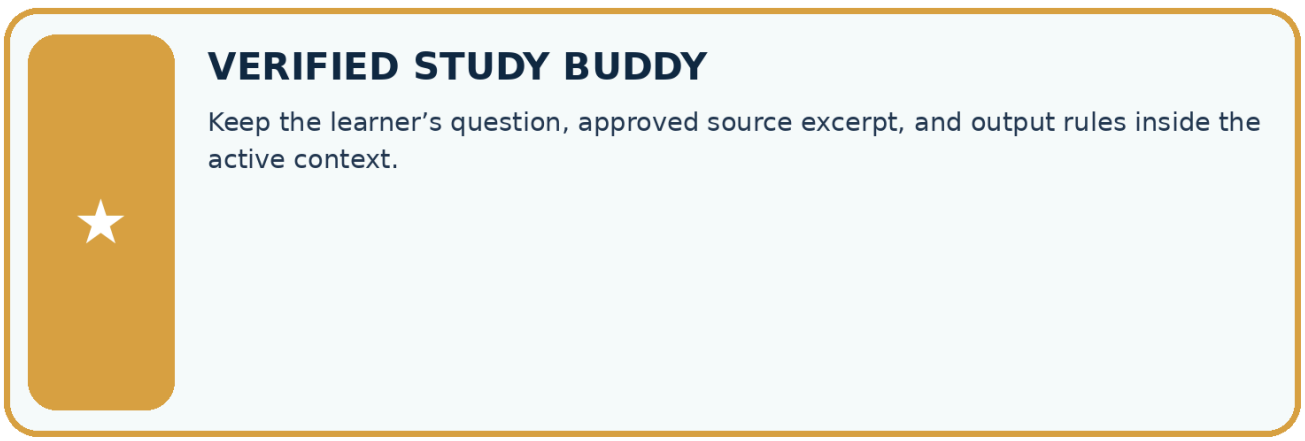
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse fluent **output** with verified truth.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Verified Study Buddy

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: show the budget and the claim-support path

> Explain what fitting evidence does and does not establish.

Explain without notes

- Close your notes. Explain this goal in your own words: Account for fixed text, **evidence** and **reserved output** under a toy budget

My deciding step, evidence and one remaining question

Plan a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
token	A unit of encoded text processed by a language model, which may be a word, part of a word or punctuation.	
tokenization	Converting text into the token units of a particular tokenizer.	
context window	The bounded amount of token information a model can process together in a particular use.	
truncation	Removing part of information, often because it exceeds a length or context limit.	

Main explanation terms

Word	Meaning
Context budget	The permitted input and output allocation under a stated model or classroom contract

Word	Meaning
Evidence selection	Choosing source material relevant to the requested claim
Reserved output	Capacity deliberately kept available for the response

Extension terms

Term	Meaning
Query	A vector used to compare available keys for the current calculation
Key	A vector used to determine compatibility with a query
Value	A vector combined using the resulting attention weights
Mask	A rule excluding specified positions from a calculation

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

1

TOKEN

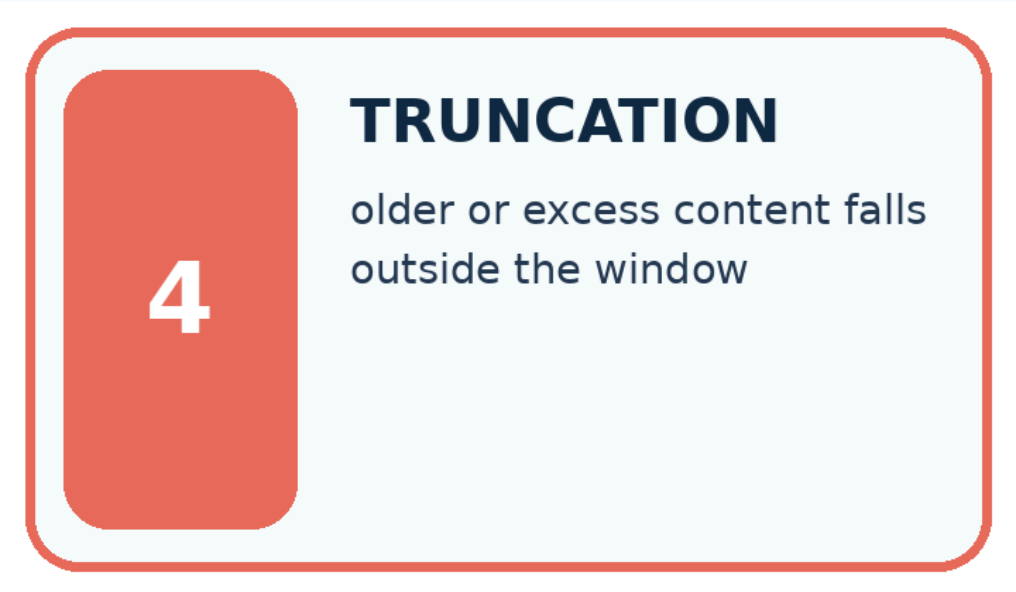
piece of text processed by a model

2

TOKENIZATION

method for splitting text into pieces





Words used in the closer explanation

Terms for Attention as weighted information selection

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

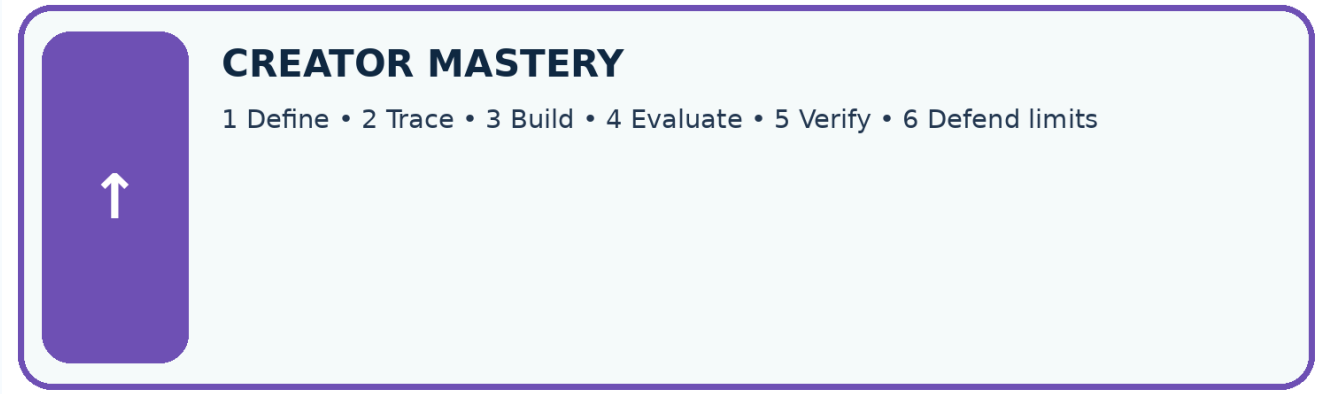
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

REFERENCE · SECTION 12

Reference · Check your explanation

> Use after practice; Budget a new evidence set independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Write or draw your reasoning and evidence.

Self-check

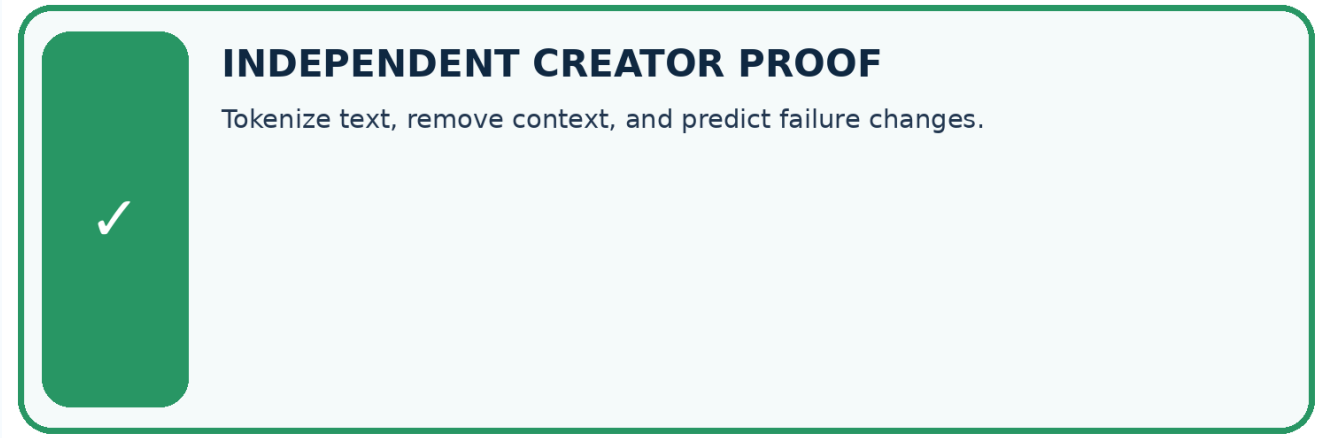
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week's explanation

Use after practice. Budget a new evidence set independently is the selected independent task.



Require budget arithmetic, current and complete support, and an explicit missing-evidence fallback.

▼ **Extra practice · The earlier complete case**

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

A toy window holds eight **token**s; prompt uses six and quotation uses three.

Does it fit? What should be preserved?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

Check after trying

Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.

Evidence or stage	Value or result
Prompt	6
Quotation	3
Capacity	8

Earlier wording, alternatives and picture transcripts

Week 3 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Nine requested tokens exceed eight spaces

A toy window holds eight **token**s; prompt uses six and quotation uses three.

Does it fit? What should be preserved?

Nine exceeds eight. Shorten or select material while preserving essential instructions and **evidence**. Real systems also have **output**-budget rules.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Account for fixed text, evidence and reserved output under a toy budget	Explain the deciding step, show a trace or test, and state one limit.
Select complete evidence for a stated question	Explain the deciding step, show a trace or test, and state one limit.
Handle insufficient or conflicting evidence explicitly	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional assistant has a small **context budget** for answering an event-capacity question.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional assistant has a small **context budget** for answering an event-capacity question.
- Which labelled step matches this detail? For this event, the declared cap is 12, with both A and B retained.
- 1. Answer with scope
- 2. Select relevant pair
- 3. Reject incomplete shortcut
- 4. Compute budget
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after trying**

After your attempt, compare with the supported explanation: Answer with scope For this event, the declared cap is 12, with both A and B retained. A **context window** limits how much information is available in one model call.

Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement. Keep the relevant source relationship visible.

Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Changed-case question

- Changed-case practice: Increase B's cost to 30 with all other costs fixed. Which conclusion is supported by the supplied facts?
- 1. A+B costs 48 and no longer fits the 45-**token evidence** capacity.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Repair and keep evidence

Review A Distribution of Possibilities (ai-3/2). Goal: Check whether a next-**token** table is a valid probability distribution. Short practice: Why is 0.625 not a broken probability sum?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

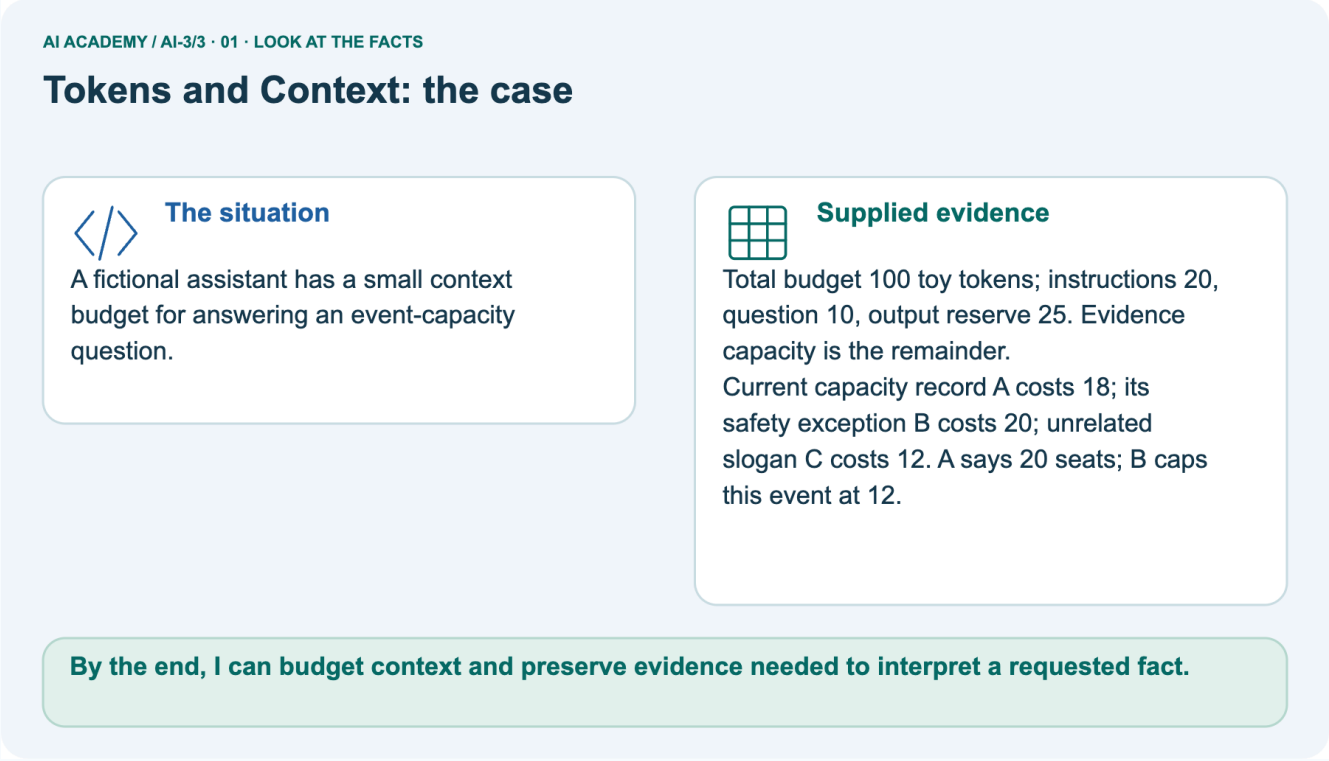
▼ **Reference · The six-picture journey**

REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Tokens and Context: the case



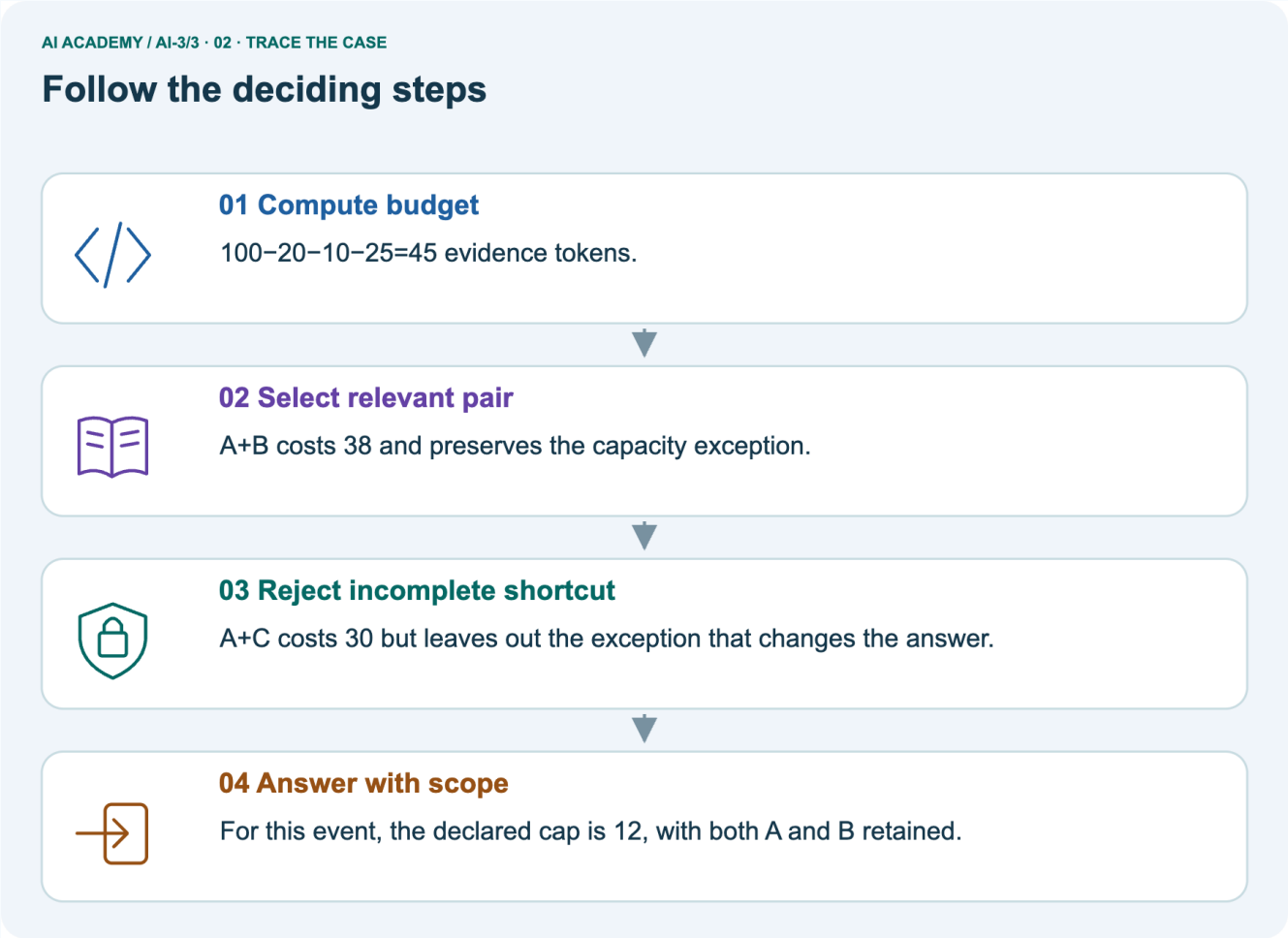
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional assistant has a small **context budget** for answering an event-capacity question. Supplied facts: Total budget 100 toy **token**s; instructions 20, question 10, **output** reserve 25. **Evidence** capacity is the remainder.

Current capacity record A costs 18; its safety exception B costs 20; unrelated slogan C costs 12. A says 20 seats; B caps this event at 12.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute budget: 100−20−10−25=45 **evidence token**s. → Select relevant pair: A+B costs 38 and preserves the capacity exception. → Reject incomplete shortcut: A+C costs 30 but leaves out the exception that changes the answer. → Answer with scope: For this event, the declared cap is 12, with both A and B retained.

Point to each numbered stage and explain which recorded **input** it uses.

03. Context budget allocations

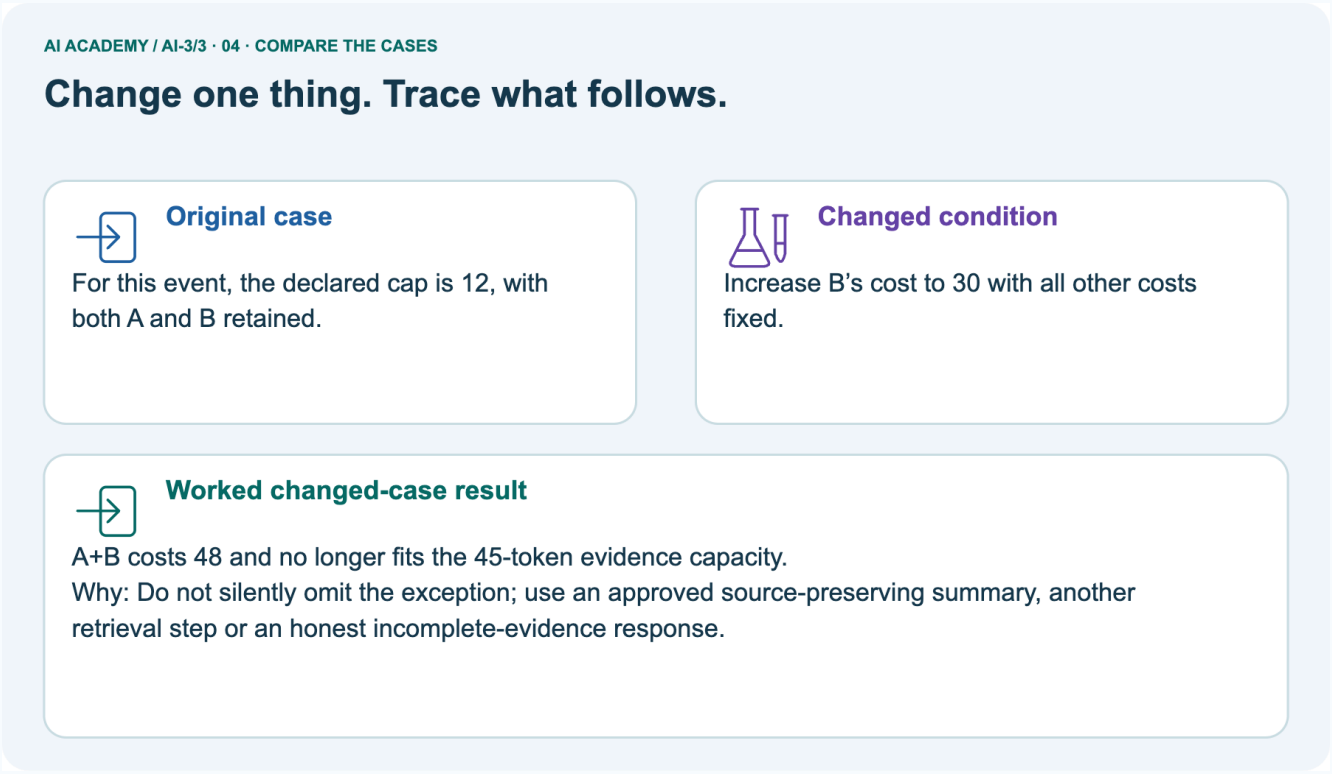


Follow the labelled picture with the matching explanation. Open the vector diagram

Context budget allocations. Instructions: Toy **token**s = 20. Question: Toy **token**s = 10. **Output** reserve: Toy **token**s = 25. **Evidence** available: Toy **token**s = 45.

Read the labels and **value**s. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

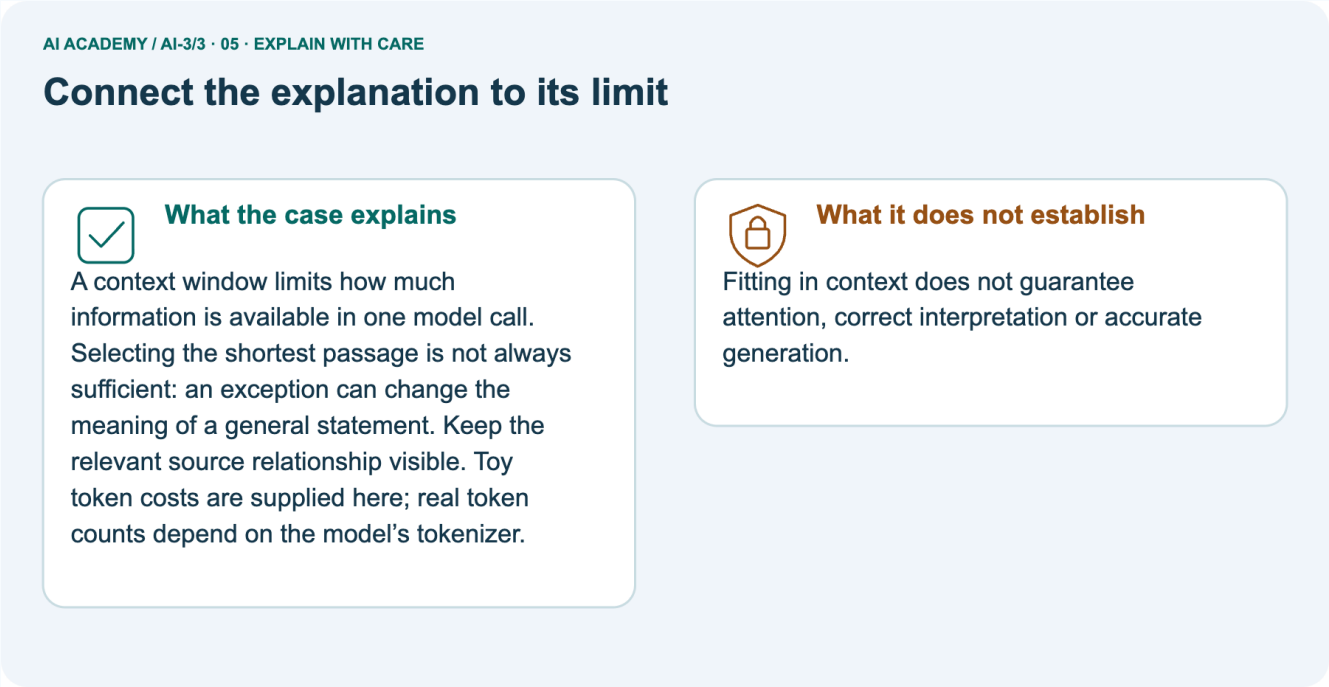
Original case: For this event, the declared cap is 12, with both A and B retained. Changed condition: Increase B's cost to 30 with all other costs fixed.

Expected result: A+B costs 48 and no longer fits the 45-**token evidence** capacity.

Reason: Do not silently omit the exception; use an approved source-preserving summary, another retrieval step or an honest incomplete-**evidence** response.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

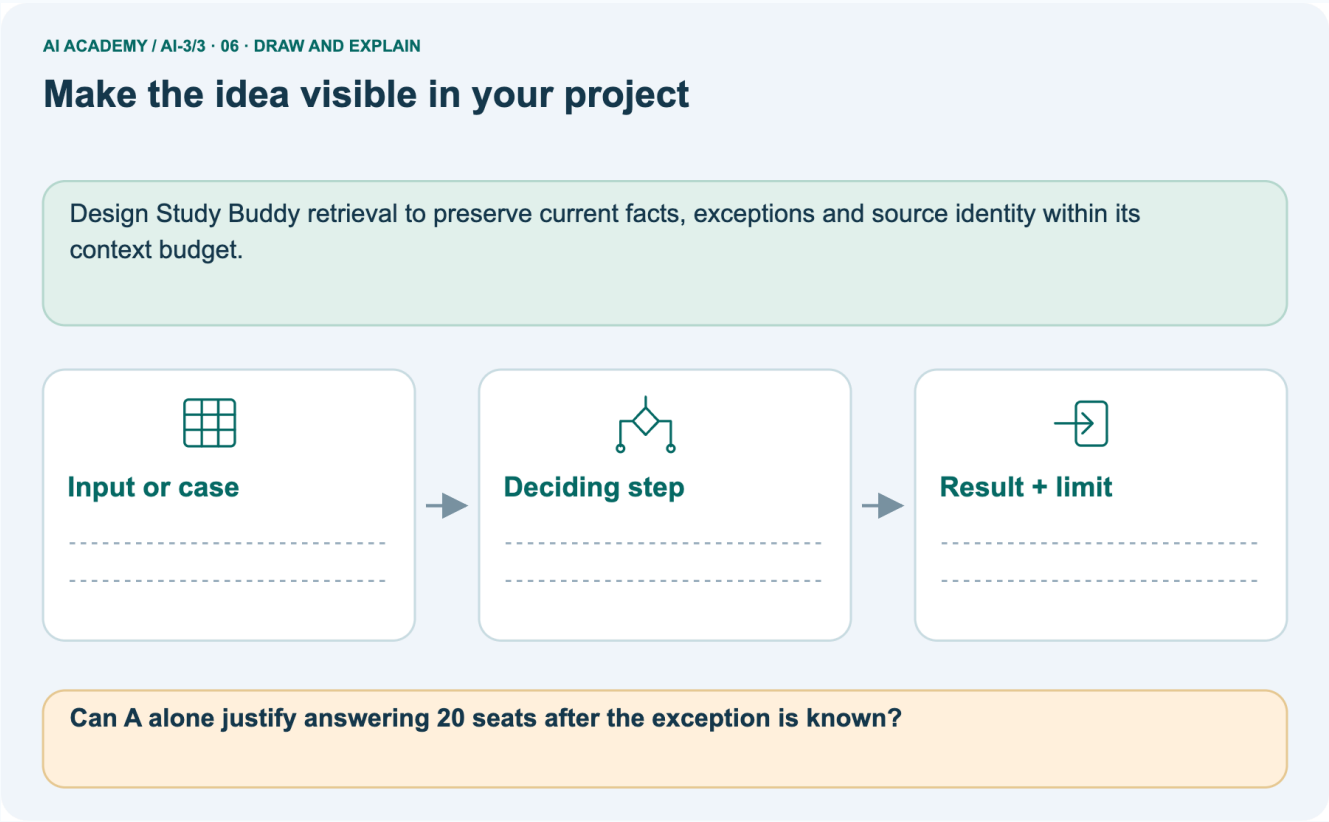
A **context window** limits how much information is available in one model call. Selecting the shortest passage is not always sufficient: an exception can change the meaning of a general statement.

Keep the relevant source relationship visible. Toy **token** costs are supplied here; real **token** counts depend on the model's **token**izer.

Limit: **Fitting** in context does not guarantee attention, correct interpretation or accurate generation.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Design Study Buddy retrieval to preserve current facts, exceptions and source identity within its **context budget**. Draw your **input**, deciding step, result and limit.

Supported question: Can A alone justify answering 20 seats after the exception is known?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Tokens and Context: the case

02. Follow the deciding steps

03. Context budget allocations

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.